Attenuation of 2.4 GHz Wi-Fi Signals by Common Building Materials: An ESP32 RSSI Measurement Study

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Abstract

Wi-Fi performance inside a building depends heavily on what the radio waves have to pass through, yet the numbers are rarely available for the materials used in ordinary homes in South Asia. This study measured how much a set of common building materials weakens a 2.4 GHz Wi-Fi link. Two ESP32 DevKit V1 boards formed a fixed-channel access point and station 1.83 m apart at a height of 0.91 m. For each condition the station recorded 30 received signal strength indicator (RSSI) readings at one-second intervals, together with the RSSI seen at the access point, the UDP round-trip time and the packet loss. Seven conditions were analysed: an unobstructed baseline (mean −66.63 dBm), 3 mm glass, 9 mm gypsum board, 12 mm plywood, 25 mm solid wood, a 50 mm concrete slab and a 254 mm (10-inch) wall. The 10-inch wall was the only material to cause a large loss, 12.9 dB (95% CI ±1.0 dB), which corresponds to roughly 95% of the received power. Solid wood caused a small but consistent loss of 2.0 dB. Glass, gypsum board and plywood could not be distinguished from the baseline within the run-to-run variation of about 1 to 2 dB. The concrete slab produced an apparent *gain* of 5.9 dB, which cannot be true attenuation and is discussed as a likely multipath or geometry effect. The brick run was excluded because its log file duplicated the baseline. No packet was lost and the round-trip time stayed near 6 ms in every condition, so a single wall was not enough on its own to degrade a short, idle link. The results, and the limits of a one-run-per-condition design, are discussed.

Keywords: Wi-Fi, 2.4 GHz, RSSI, signal attenuation, building materials, ESP32, indoor propagation

1. Introduction

The problem that started this study was an ordinary one. In the last room of my house, the ping in an online game (Free Fire) regularly rose to between 150 and 300 ms. Standing nearer to the router, or in another room at about the same distance from the router but with glass walls, the ping was noticeably lower. My first assumption was that the whole network was at fault. After reading about indoor radio propagation I began to suspect that the walls between the router and that room were weakening the signal, so that the connection in the last room was simply a weaker radio link.

That suspicion is easy to state and harder to test. A phone or laptop reports signal strength, but it cannot isolate one wall or one material. A controlled experiment can: keep the transmitter, the receiver, the distance and the radio settings fixed, and change only the object placed between them. Low-cost microcontrollers make that possible. The ESP32 has a built-in 2.4 GHz Wi-Fi radio that reports the received signal strength of every connection, and two boards cost very little.

The aim of this study was therefore to measure, with the same hardware and geometry for every test, how much a selection of common building materials attenuates a 2.4 GHz Wi-Fi signal, and to see whether the weaker signal also shows up as a longer round-trip time or lost packets. The specific questions were:

1. How much weaker is the received signal when glass, gypsum board, plywood, solid wood, a concrete slab, brick or a 10-inch wall is placed between the two nodes?
2. Does the loss grow with material thickness, as simple models suggest?

3. Do the measured losses change the round-trip time or the packet loss of a short link?

2. Background

2.1 Received signal strength and attenuation

Wi-Fi in the 2.4 GHz band follows the IEEE 802.11 standard [1]. Each receiver estimates the power of the frames it receives and reports it as a received signal strength indicator, RSSI. In the ESP32 software the value is returned in whole decibel-milliwatts (dBm) [2]. RSSI is a relative indicator: its absolute calibration differs between chips and antenna designs, so the differences between two measurements made with the same hardware are more meaningful than the values themselves.

In this paper the attenuation caused by a material is defined as the drop in mean RSSI relative to an unobstructed baseline measured with the same nodes and geometry:

$$A = R_0 - R_m \tag{1}$$

where R_0 is the mean RSSI of the baseline run and R_m is the mean RSSI with material m in the path. A positive A means signal was lost. A ratio of powers follows from the decibel scale: a loss of 10 dB leaves one tenth of the power and 13 dB leaves roughly one twentieth.

2.2 Why materials matter

In free space the power at distance d falls according to the Friis transmission formula [3]. Expressed as a path loss it is

$$\mathrm{FSPL\ (dB)} = 20 \log_{10}(4\pi d/\lambda) \tag{2}$$

with wavelength λ. On channel 6 (2437 MHz) λ is 12.3 cm, and at the 1.83 m used here the free-space loss is about 45.4 dB. Inside buildings the signal also passes through, reflects from and diffracts around walls, floors and furniture, so measured losses are larger and fluctuate with position [4], [5]. A material weakens a signal by reflecting part of it and by absorbing part of it as heat; denser, wetter and more conductive materials generally do both more strongly. Reference data for building materials, including their dependence on frequency and moisture, are collected in ITU-R Recommendation P.2040 [6] and in the NIST measurements of construction materials [7]. Because the quantities involved are measured per material and are sensitive to moisture and construction, a local measurement with locally available materials is still useful.

Signal strength fluctuates even when nothing changes, because reflections from nearby surfaces add to the direct signal with different phases (multipath fading) and because people and objects move. This variation is well known from indoor RSSI studies [8] and sets the smallest loss that a single measurement run can reliably resolve.

3. Materials and Methods

3.1 Hardware

Two ESP32 DevKit V1 boards were used. One board acted as the transmitter and Wi-Fi access point (AP). It was set to a fixed channel (channel 6, 2437 MHz), a fixed SSID, a single permitted client and the highest transmit-power setting of the library (19.5 dBm), which was kept unchanged for all tests. The second board acted as the receiver and station (STA). Wi-Fi power saving was switched off on the receiver to keep RSSI and latency steady. Both boards were programmed in the Arduino IDE, and the receiver's serial output was logged on a computer. The boards carry a printed-trace antenna, which is not isotropic; this matters for the interpretation of the results (Section 5.3).

3.2 Firmware and quantities recorded

The transmitter also ran a small UDP echo server. Once per second the receiver sent a numbered UDP packet to the access point. The access point replied with the same sequence number together with the RSSI it had measured for the receiver's packet. For each of 30 readings the receiver then logged:

- **RSSI at the receiver** (dBm), read from the Wi-Fi driver at the start of each one-second interval. This is the main response variable.
- **RSSI at the access point** (dBm), returned in the echo. It gives a second, independent estimate of the link in the opposite direction.
- **Round-trip time** (RTT, ms), measured from sending the packet to receiving the matching echo, with an 800 ms timeout.
- **Packet status**, from which packet loss was computed.

The sketch stops after exactly 30 readings and prints the sample mean, sample standard deviation ($n - 1$), minimum, maximum, median, packet loss and RTT statistics to the serial monitor. The serial output was copied into one .csv file per condition.

3.3 Geometry

The two boards were mounted on wooden stands, 6 ft (1.83 m) apart and 3 ft (0.91 m) above the floor, with the same placement for every condition. Each specimen was held upright between them, 3 ft (0.91 m) from each board, so that it crossed the line of sight at its midpoint. The first Fresnel zone of the link has a radius of about 0.24 m at the midpoint, which is smaller than the specimens, so little signal should have bypassed them by diffraction around the edges. The tests were carried out in a single indoor room with painted walls, a hard floor and a window (Figure 1). Nearby Wi-Fi networks and other sources of 2.4 GHz interference were not surveyed.



Figure 1. The measurement room. The wooden stands that carried the two ESP32 boards are visible, with the specimens leaned against the wall at the back of the room.

3.4 Specimens

Seven materials were prepared, listed in Table 1. Thicknesses were measured with a ruler and recorded in the field notes. The thinnest specimen was the 3 mm glass pane and the thickest was the 254 mm wall. Photographs of four of the specimens are shown in Figure 2.

Table 1. Test conditions and specimens.

Condition	Thickness	Description (from field notes and photographs)	Status in analysis
No obstruction	—	Open line of sight between the two nodes	Baseline
Glass	3 mm	Single pane in a wooden frame	Analysed
Gypsum board	9 mm	Patterned gypsum board panel	Analysed
Plywood	12 mm	Plywood sheet	Analysed
Solid wood	2.5 cm (25 mm)	Solid timber board	Analysed
Concrete slab	5 cm (50 mm)	Cast slab, no steel reinforcement	Analysed
Brick	7.7 cm (77 mm)	Bricks stacked to form a screen	Excluded (Section 3.6)
10-inch wall	10 in (254 mm)	Wall of the building	Analysed



Figure 2. Examples of the specimens. Top row: the stacked bricks and the framed glass pane. Bottom row: the gypsum board and the wooden board. The paper slips on each specimen record its dimensions.

3.5 Procedure

For each condition the specimen was placed between the boards, the receiver was reset, and it connected to the access point and took 30 readings, one every second. The unobstructed run was recorded with nothing between the boards and serves as the baseline in Equation 1. The transmitter stayed in the same position and orientation throughout.

3.6 Data handling

The mean and standard deviation of each condition were recomputed from the raw readings, and they agree with the summaries printed by the firmware. Three issues in the log files were handled as follows.

- **Metadata fields.** The thickness, distance and height printed in the .csv files are constants in the firmware, and they were not edited for every run (for example, they read 0 mm, 2 m and 5 ft), and some files still carry the placeholder label “Baseline”. The true values are those in Sections 3.3 and 3.4, taken from the field notes.
- **Brick run.** The brick file contained three repeated rows (sample numbers 8 to 10). After removing them, its 30 readings of receiver RSSI, access-point RSSI and RTT were identical, value for value and timestamp for timestamp, to those of the unobstructed run. Two independent runs cannot be identical, so the file is a copy and the brick condition was **excluded** from the

analysis rather than reported with a borrowed result. The unobstructed run was kept as the baseline.

- **Missing fields.** A few rows (one per file for glass, plywood, wood, concrete and gypsum) have empty or mistyped metadata cells. The signal values in those rows are complete and were kept.

3.7 Statistical analysis

For each condition the mean, standard deviation, median and range of the 30 receiver RSSI readings were computed. The attenuation (Equation 1) was reported with a 95% confidence interval for the difference of means (Welch's method), together with the Welch *t*-test *p*-value and the Mann–Whitney *U* test as a check that does not assume normality. Because consecutive readings from the same run are correlated, the lag-1 autocorrelation and an approximate effective sample size, $n(1 - r)/(1 + r)$, are given in Appendix A. The *p*-values should be read as indicative where the autocorrelation is high. The differences among all seven conditions were also tested with a one-way ANOVA and a Kruskal–Wallis test. Analyses were done in Python (NumPy and SciPy).

4. Results

4.1 Received signal strength

The unobstructed baseline had a mean RSSI of −66.63 dBm (SD 1.56 dB, range −70 to −64 dBm). The receiver RSSI differed significantly among the seven conditions (one-way ANOVA, $F = 583.5$, $p < 0.001$; Kruskal–Wallis, $p < 0.001$). Table 2 gives the results for each condition, and Figures 3 to 5 show them.

Table 2. Receiver RSSI (30 readings per condition) and attenuation relative to the unobstructed baseline†. Positive attenuation is signal lost; negative is signal gained.

Condition	Thick. (mm)	Mean RSSI (dBm)	SD (dB)	Median (dBm)	Min / Max (dBm)	Atten. (dB)	95% CI	Welch *p*
No obstruction†	0	−66.63	1.56	−66.5	−70/−64	—	—	—
Glass	3	−66.83	0.99	−67.0	−70/−65	+0.20	±0.68	0.556
Gypsum board	9	−65.87	1.33	−66.0	−70/−64	−0.77	±0.75	0.046
Plywood	12	−67.43	0.90	−67.5	−69/−66	+0.80	±0.66	0.019
Solid wood	25	−68.63	0.67	−69.0	−70/−67	+2.00	±0.63	<0.001
Concrete slab	50	−60.77	0.50	−61.0	−62/−60	−5.87	±0.61	<0.001
10-inch wall	254	−79.50	2.21	−80.0	−84/−74	+12.87	±0.99	<0.001

† Reference condition. Confidence intervals and *p*-values are for the difference from the baseline.

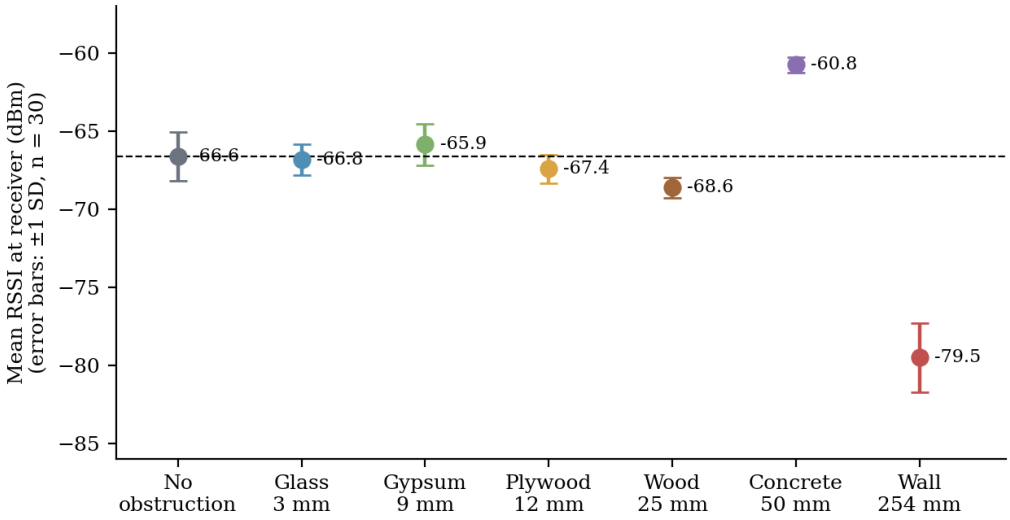


Figure 3. Mean receiver RSSI for each condition (error bars: ±1 SD, n = 30). The dashed line marks the baseline mean.

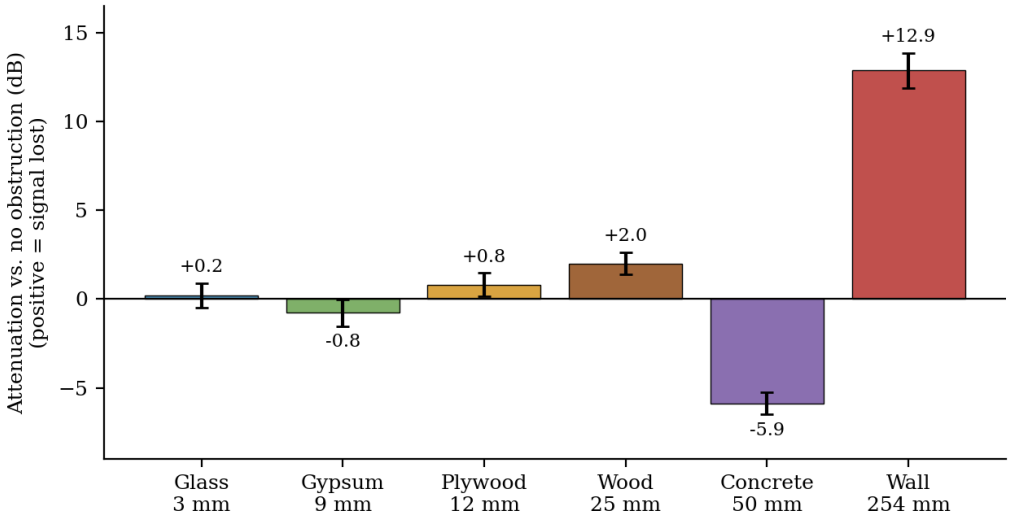


Figure 4. Attenuation relative to the unobstructed baseline, with 95% confidence intervals for the difference. Positive values mean signal was lost.

The results fall into four groups:

- **The 10-inch wall** was the only large attenuator. The mean RSSI fell to −79.50 dBm, a loss of 12.87 dB (95% CI ±0.99 dB), which is about a 19-fold reduction in received power. The run also had the largest spread (SD 2.21 dB) and drifted downward during the 30 seconds, from a mean of −75.6 dBm over the first five readings to −81.2 dBm over the last five (Figure 5). Depending on which part of the run is used, the loss is therefore between about 9 and 15 dB.
- **Solid wood (25 mm)** reduced the signal by 2.00 dB (±0.63 dB, $p < 0.001$). The loss is small, but wood also had the steadiest signal (SD 0.67 dB), and the access-point side shows a loss in the same direction (1.47 dB: mean −71.73 dBm against −70.27 dBm for the baseline, Table 3).
- **Glass, gypsum board and plywood** gave results close to the baseline: +0.20 dB for 3 mm glass ($p = 0.56$), −0.77 dB for 9 mm gypsum board ($p = 0.046$, a slightly *stronger* signal than the baseline) and +0.80 dB for 12 mm plywood ($p = 0.019$). The plywood result is the weakest of the three: its readings are strongly autocorrelated (effective n about 6.5), and the access-point side shows a gain of 0.87 dB (−69.40 dBm against −70.27 dBm) instead of a loss. None of these three is treated as a reliable attenuation.

- **The 50 mm concrete slab** gave a mean RSSI of −60.77 dBm, which is 5.87 dB *stronger* than the baseline, with the smallest spread of any run (SD 0.50 dB). The access-point side agrees (a gain of 5.43 dB: −64.83 dBm against −70.27 dBm), so the effect is a real change in the measured link and not a reading error in one board. A slab cannot add power, so this is not attenuation; Section 5.2 discusses it.

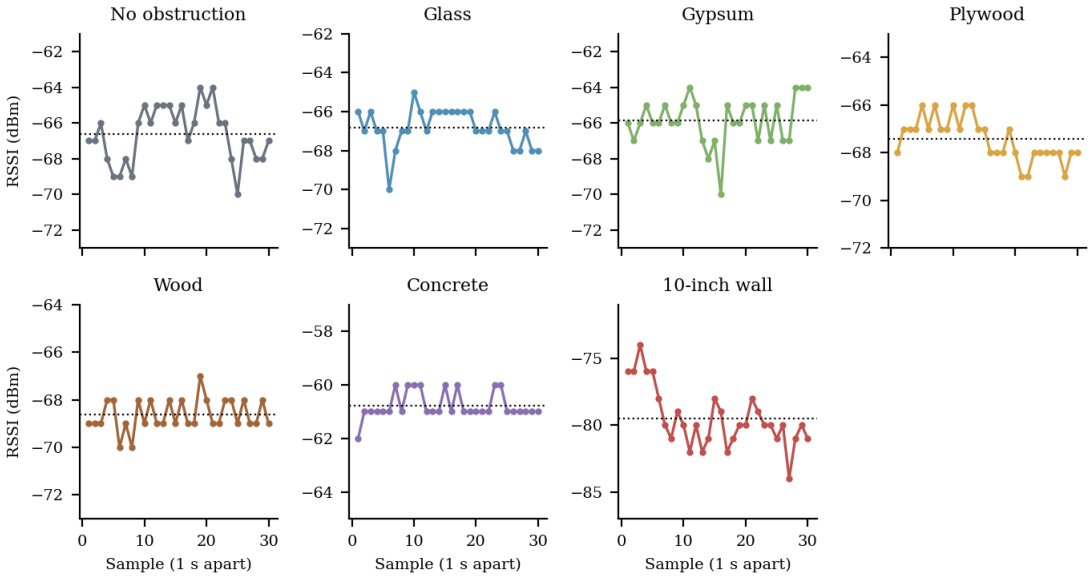


Figure 5. Receiver RSSI for each of the 30 readings in every condition. The dotted line is the mean of each run. The vertical scale differs between panels.

4.2 Access-point RSSI, latency and packet loss

Table 3 gives the RSSI measured at the access point and the link-level results. The access-point RSSI follows the receiver RSSI closely (Pearson r = 0.93 over all readings), and it shows the same main pattern: a large loss for the wall, a large gain for the concrete slab and a small loss for wood. For glass, gypsum board and plywood the two sides differ by less than 1 dB and not always in the same direction, which again points to no reliable effect. It is lower than the receiver value by about 3 to 4 dB in the baseline, which is a property of the two radios and antennas, not of the materials.

Table 3. Access-point RSSI, round-trip time and packet loss for each condition (30 packets per condition).

Condition	AP-side RSSI (dBm, mean ± SD)	RTT (ms, mean ± SD)	RTT range (ms)	Packets lost / sent
No obstruction†	−70.27 ± 1.46	5.98 ± 2.50	4.20 – 16.22	0 / 30
Glass	−69.70 ± 1.32	6.41 ± 2.17	4.21 – 12.25	0 / 30
Gypsum board	−70.60 ± 1.22	6.01 ± 2.44	4.21 – 13.20	0 / 30
Plywood	−69.40 ± 1.43	6.94 ± 1.69	5.20 – 11.22	0 / 30
Solid wood	−71.73 ± 0.64	6.48 ± 2.08	4.20 – 13.21	0 / 30
Concrete slab	−64.83 ± 1.39	6.55 ± 2.40	4.21 – 16.26	0 / 30
10-inch wall	−80.63 ± 3.61	6.41 ± 1.79	4.20 – 11.22	0 / 30

Every one of the 210 UDP packets was answered, so packet loss was 0% in all conditions. The mean round-trip time was between 5.98 and 6.94 ms. It did not differ significantly among conditions (Kruskal–Wallis, p = 0.16) and was not correlated with RSSI across all readings (Spearman ρ = −0.04, p = 0.58). Even at a mean of −79.5 dBm behind the 10-inch wall, the round-trip time was 6.41 ms (Figure 6). The occasional readings of 10 to 16 ms in every condition are isolated spikes and do not depend on the material.

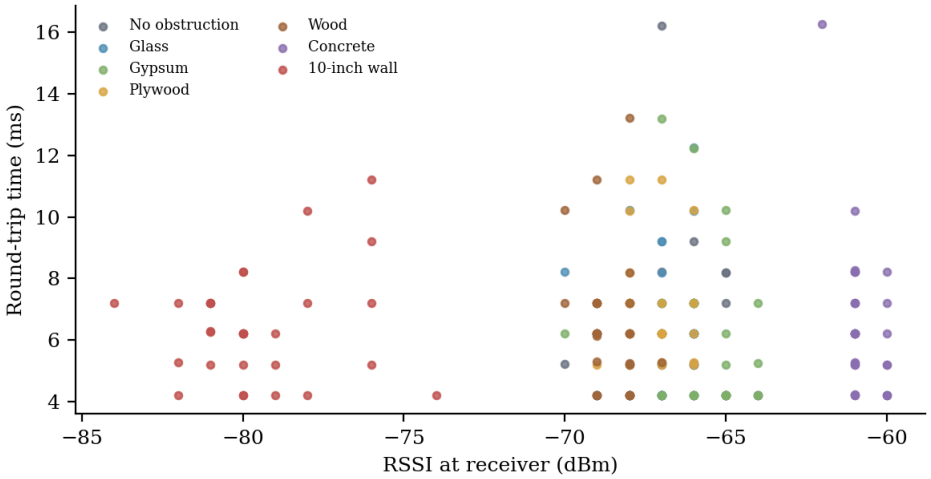


Figure 6. Round-trip time against receiver RSSI for all readings. No relationship is visible.

5. Discussion

5.1 What the measurements show

The clearest result is the 10-inch wall. Passing through it removed about 13 dB of signal, which means that only about one-twentieth of the power was left. A single wall of this kind is therefore enough to turn a strong indoor link into a weak one, and a signal that has to cross two such walls would lose of the order of 25 dB if the effect simply added. Per unit thickness, the wall (about 0.5 dB/cm) and the solid wood board (about 0.8 dB/cm) lose a similar amount, but this is a comparison of one specimen of each, and the plywood, glass and gypsum results show that readings this close to the noise cannot support a per-centimetre rule. The data do not show a clean increase of loss with thickness among the thin and medium specimens; the only clear increase is from the thin materials to the thick wall.

Glass, gypsum board and plywood were effectively transparent at this precision. That agrees with the general expectation that thin, dry, low-density partitions attenuate 2.4 GHz only a little [6], [7], and it fits the observation that started this study: the room with glass walls gave a better connection than the room behind masonry. The data cannot say whether glass is *better* than other thin partitions, only that three millimetres of glass did not measurably weaken the link.

5.2 The concrete result

The concrete slab gave a signal 5.9 dB stronger than the baseline, in both directions of the link. This should not be reported as negative attenuation of the slab. The more likely explanation is in the geometry. At 2.4 GHz the wavelength is only 12 cm, and the received power is the sum of the direct wave and every reflection. A dense slab 0.91 m from each board can act as a strong reflector, and when the reflected wave adds in phase with the direct wave, the total can exceed the free-path value, while a different position or angle can add out of phase and cancel. The unobstructed baseline may itself sit in a partial fade, which would make every other condition look better relative to it. Small changes in the position or angle of the boards between runs would have a similar effect, because their printed antennas are not isotropic.

The gypsum result (0.8 dB stronger than the baseline) and the concrete result together show that run-to-run differences of the order of 1 dB to several dB can arise from the arrangement alone. This sets a practical noise floor for a design that uses one run per condition: effects smaller than about 2 dB, which includes glass, gypsum board and plywood here, cannot be separated from it. The present data cannot

confirm the multipath explanation; it is a hypothesis to test by repeating the concrete run with the boards re-positioned and rotated.

5.3 What the latency result means for the original problem

My original problem was a ping of 150 to 300 ms. In these experiments a 13 dB loss lowered the signal to about −80 dBm but did not change the round-trip time (about 6 ms) or cause any lost packet. At 1.83 m and with one packet per second, the link stayed well inside the range in which Wi-Fi works normally. So this study does not show that one wall alone causes a high ping. What it shows is that a wall of this type costs about 13 dB, which is a large part of the margin of a typical home link. In my house the last room is further from the router than the 1.83 m used here, and the signal probably crosses more than one wall. Together these could push the link to a weak level, at which a router lowers its data rate, retransmits more frames and is more affected by other networks on the same channel. That chain was not tested here. Latency under load was not measured and I did not measure the signal between the router and the last room, so this remains an explanation to be tested, not a finding.

5.4 Limitations

- **One specimen and one run per condition.** There is no replication, so run-to-run variation cannot be separated from material effects. Replicated runs in randomised order are the main improvement needed.
- **One baseline.** All attenuation values depend on a single unobstructed run, whose own value may be shifted by multipath.
- **Short, correlated runs.** Each run lasted 30 s, and consecutive readings are correlated (Appendix A), so the effective sample is smaller than 30 and the *p*-values are optimistic for some conditions.
- **RSSI resolution and calibration.** RSSI is reported in whole dB and is not an absolute power calibration. The measured baseline (about −66.6 dBm) is about 40 dB below the 19.5 dBm transmit setting minus the 45.4 dB free-space loss. The gap is attributable to antenna efficiency and pattern, the chip's reporting offset and indoor effects. Only differences within this study are interpreted.
- **Uncontrolled environment.** Interference from other Wi-Fi networks, moving people and the moisture content of the specimens were not controlled or recorded. The identity of the specimens as pure materials was not independently verified.
- **Missing brick data.** Brick, one of the most common building materials in the region, is absent from the results.
- **Frequency.** Only the 2.4 GHz band was tested. The original ESP32 does not support 5 GHz, and losses at 5 GHz are generally higher.

6. Conclusion

Using two ESP32 boards, seven conditions were measured with 30 RSSI readings each. The main conclusions are:

1. A 10-inch (254 mm) wall attenuated a 2.4 GHz Wi-Fi link by 12.9 dB (95% CI ±1.0 dB), about 95% of the received power. It is the only material in this study with a large and unambiguous effect.
2. 25 mm solid wood attenuated the link by a small 2.0 dB. 3 mm glass, 9 mm gypsum board and 12 mm plywood were indistinguishable from the unobstructed baseline within the roughly 1 to 2 dB variation of single runs.

3. The concrete slab produced an apparent gain of 5.9 dB. This is a measurement artefact, most likely multipath or a change in geometry, and not a property of concrete. The concrete result needs repetition before any conclusion about concrete is drawn.
4. The brick measurement was invalid (its log duplicated the baseline) and must be repeated.
5. Round-trip time (about 6 ms) and packet loss (0%) did not change with any material on this short, idle link. The effect of the 13 dB wall loss on latency in a real home is a hypothesis for further work.

The experiment is a small, honest first step, and its main value is the procedure: fixed hardware, fixed channel and geometry, a documented analysis and a clear statement of what each result can and cannot support.

Future work

- Repeat every condition at least five times, in randomised order, with the boards re-positioned between repeats and a new baseline for each block.
- Repeat the brick run, and repeat the concrete run with the boards rotated and moved in small steps to test the multipath explanation.
- Measure through one, two and three walls, and at several distances, along the actual path from the router to the last room of the house.
- Measure round-trip time and throughput under load, not only at one packet per second, to relate signal loss to the gaming ping that motivated this work.
- Survey the channel for interference and record the moisture and exact size of each specimen.

Acknowledgements

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Appendix A. Additional statistics

Table A1 gives the standard error of the mean and the measures used to judge how far the *p*-values can be trusted: the lag-1 autocorrelation of the 30 readings, the linear trend within each run, the approximate effective sample size, and effect sizes. Autocorrelation is highest for the baseline, plywood and wall runs (about 0.6), where the signal wandered slowly during the run.

Table A1. Additional statistics for the receiver RSSI of each condition.

Condition	SE of mean (dB)	Lag-1 autocorr.	Linear trend (dB/s)	Approx. effective *n*	Cohen's *d*	Mann–Whitney *p*
No obstruction†	0.29	0.60	+0.007	7.4	—	—
Glass	0.18	0.43	−0.020	11.9	0.15	0.502
Gypsum board	0.24	0.19	+0.023	20.5	−0.53	0.050
Plywood	0.16	0.64	−0.067	6.5	0.63	0.021
Solid wood	0.12	−0.17	+0.015	30.0	1.66	<0.001
Concrete slab	0.09	0.15	+0.003	22.3	−5.05	<0.001
10-inch wall	0.40	0.63	−0.155	6.9	6.72	<0.001

Appendix B. Firmware and data

The transmitter sketch (access point and UDP echo server) and the receiver sketch (30 readings, one per second, then summary statistics) were written for the Arduino IDE and the ESP32 DevKit V1. The settings that were identical for every test were: channel 6, transmit power 19.5 dBm, one permitted client, receiver power saving off, 30 readings at 1 s spacing and an 800 ms echo timeout. The raw readings of every run are in the accompanying .csv files, one per condition.